Mixing Length Turbulence Model

Problem

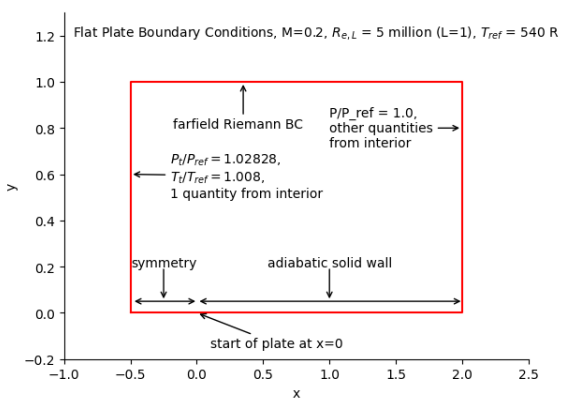
The objective of this study is to model the turbulent boundary layer over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the mixing length turbulence model. This simulation will validate the CFD results against theoretical predictions by comparing:

- Wall skin friction coefficient (C_f) at different momentum-thickness Reynolds numbers ($4000 < R_{e,\theta} < 13000$).
- The law of the wall at $R_{e,\theta} = 10000$ to verify the near-wall velocity distribution.
- Eddy viscosity profiles at specific downstream locations (x=0.2, x=0.4, and x=0.6 m).

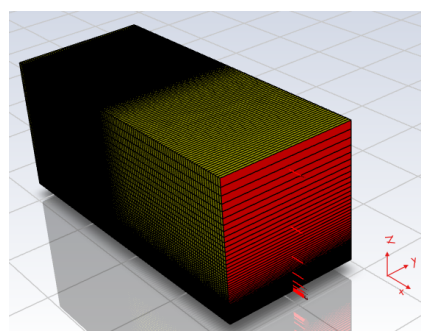
The aim is to evaluate the accuracy of the mixing length turbulence model and its ability to capture essential flow characteristics in a turbulent boundary layer.

Solution

The study of boundary layer flow over a flat plate is fundamental in fluid mechanics, particularly in understanding transition from laminar to turbulent flow. Boundary conditions along with 2D representation of the plate is presented on the Figure 1a.



(a) 2D flat surface with boundary conditions.



(b) Mesh representation in Fluent.

Setup

At the inlet, a uniform free-stream velocity U_∞ is prescribed to simulate the flow approaching the flat plate. Turbulence intensity is set based on the Reynolds number, ensuring accurate boundary layer development. A pressure outlet boundary condition is applied at the downstream edge of the computational domain, with the pressure set to zero gauge pressure. This allows for a smooth exit of the fluid, reducing artificial reflections. The flat plate surface is modeled using a no-slip

boundary condition, where both the streamwise and normal velocities are set to zero. This enforces the correct shear stress distribution necessary for boundary layer formation. A far-field boundary condition is applied to the upper edge of the domain. This allows the flow to maintain free-stream conditions, with the velocity set to U_∞ and pressure set to ambient values.

A CFD solver is used to simulate the flow with the following settings:

- Turbulence Model: Mixing Length Model
- Grid: Structured mesh with refinement near the wall, 273×193
- Solver: Pressure-based, steady-state
- Post-processing: Skin friction coefficient, the law of the wall and eddy viscosity plots

Set of equations used in post-processing is presented below.

$$Re_\theta = \frac{\rho_\infty U_\infty \theta}{\mu_\infty} \tag{1}$$

$$\theta = \int_0^\infty \frac{\rho}{\rho_\infty} \frac{u}{U_\infty} \left(1 - \frac{u}{U_\infty}\right) dy \tag{2}$$

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{3}$$

$$\tau_w = \mu_w \left(\frac{\partial u}{\partial y}\right)_w \tag{4}$$

$$u^+ = \frac{u}{v^*} = \frac{u}{\sqrt{\tau_w/\rho_w}} \tag{5}$$

$$y^+ = \frac{y v^*}{\nu_w} = \frac{y\sqrt{\tau_w/\rho_w}}{\nu_w} \tag{6}$$

Mixing Length Model

The mixing length model is a simple eddy viscosity model used for describing turbulence in shear flows. It assumes that turbulent eddies behave like molecules in a laminar fluid, transferring momentum over a characteristic length called the mixing length (l). Unlike modern two-equation models (e.g., $k-\varepsilon$, $k-\omega$), it does not solve transport equations for turbulence quantities but instead expresses eddy viscosity as a function of velocity gradients.

The turbulent eddy viscosity is given by

$$\nu_t = l^2 \left|\frac{dU}{dy}\right|, \tag{7}$$

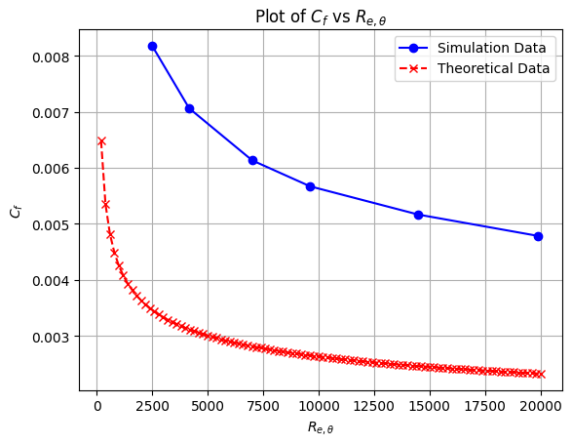
where l is mixing length, U is the mean field velocity and y is the distance from the wall. This mode assumes that mixing length varies with position, typically using empirical formula:

$$l = \kappa y, \tag{8}$$

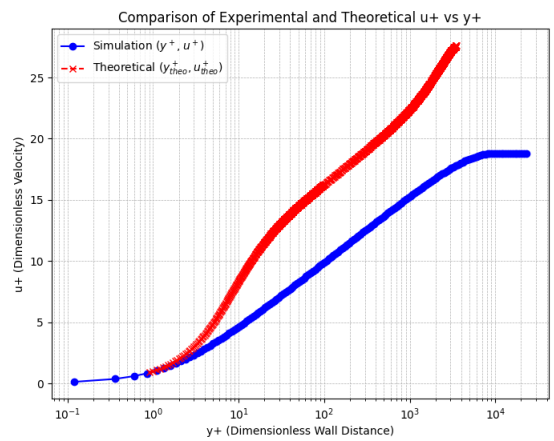
where $\kappa \approx 0.41$, and y is the distance from the wall.

Results & Discussion

Distributions of the wall skin friction coefficient C_f over different Reynolds numbers against theoretical values for this case are presented on Fig. 2a. The Law of the Wall for different turbulence models against theoretical values is presented on the Fig. 2b.



(a) The wall skin friction coefficient over the Reylonds number for the range $4000 < R_e < 13000$ for turbulence mixing length model.



(b) The law of the wall distributions for mixing length model (R_e=10000).

It can be seen on Fig. 2a that mixing length model in this case overestimates/over-predicts the wall friction coefficient. One of the reasons is that mixing length model does not account for the viscous sublayer in turbulent boundary layers. It assumes that the turbulent eddy viscosity is always proportional to the velocity gradient, even very close to the wall. in reality, near-wall turbulence is damped by molecular viscosity, leading to a low turbulence production region. By ignoring this damping effect, the model predicts too much turbulence near the wall, leading to higher shear stress and an overestimated C_f.

In the same spirit, the law of the wall is calculated and plotted against theoretical values for $R_{e,\theta} =$ 10000. Result is depicted on Fig. 2b. In this case, that Reynolds number shows around surface line x=0.6. It can be seen that only in viscous sublayer, the mixing length model and theoretical predictions are in good accord. Since turbulence is weak in the viscous sublayer, assuming a constant or very small mixing length does not introduce significant errors. The model fails in the viscous sublayer, where eddy viscosity should be much smaller. Since the model does not reduce eddy viscosity properly at low y, it predicts excessive momentum transfer, increasing wall shear stress and friction.

Eddy viscosity is plotted against the distance from the wall and for different x positions. Results are presented on Fig. 3. In a viscous sublayer, the eddy viscosity $\nu_t \approx 0$, since the molecular viscosity dominates and the turbulence is negligible. After that, the mixing length gradually increases. The gradient of mean field velocity is still large, leading to rising eddy viscosity. The shape of ν_t begins to rise steeply. In log-law layer the ν_t reaches it maximum as some intermediate value y. Beyond this maximum, velocity gradient starts decreasing due to the slower velocity variation far from the wall. This causes the ν_t to gradually decrease in the outer layer. In the outer layer, the l saturates

ar a value close to the boundary layer thickness. The velocity gradient continues decreasing and eddy viscosity starts decreasing. Reaching the minimum, the eddy viscosity starts increasing again due to the continued development of the turbulent boundary layer.

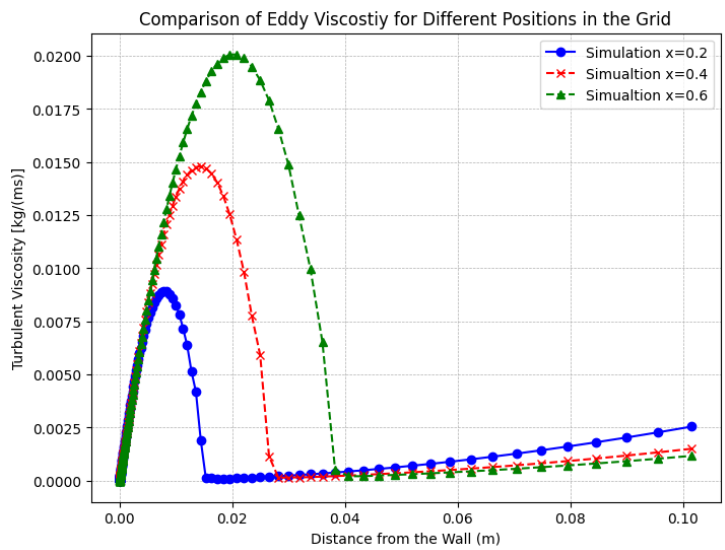


Figure 3: Eddy viscosity as a function of z-axis, distance from the wall for different positions at the surface x=0.2, 0.4 and 0.6, respectively.

Eddy viscosity distribution is plotted for three different stream-wise positions x=0.2, 0.4 and 0.6. It can be observed that for higher value of x, the eddy viscosity is larger, and it is because of the development of the turbulent boundary layer along the flow direction. As the fluid flows over a surface, the boundary layer thickens with increasing x. More turbulence is generated as the boundary layer grows, leading to higher turbulence intensity. Since eddy viscosity is proportional to turbulence intensity, it increases with x.

Conclusion

The results show that the mixing length model overpredicts the wall friction coefficient due to its inability to account for the viscous sublayer properly. This leads to excessive turbulence near the wall, increasing shear stress and overestimating C_f. The law of the wall comparison indicates that the model performs well only in the viscous sublayer but deviates significantly in the logarithmic and outer layers.

The eddy viscosity distribution confirms that turbulence intensity increases downstream, with larger x positions exhibiting higher eddy viscosity due to the thickening boundary layer. The maximum eddy viscosity occurs in the logarithmic layer before decreasing in the outer region, following expected boundary layer behavior. While the mixing length model captures general trends, its limitations in near-wall turbulence modeling lead to inaccuracies in predicting wall shear stress and velocity profiles.